

From Discrete Model to Continuum Limit

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* “paper” in the following content refers to [Anguige's paper](#)

Our original (discrete) conservation equation is the following: (equation 4 in the paper)

$$\frac{\partial \rho_i}{\partial t} = \frac{J_{i-\frac{1}{2}} - J_{i+\frac{1}{2}}}{h}$$

$J_{i+\frac{1}{2}}$ is the net flux of particles from x_i to x_{i+1} , where (equation 7 in the paper)

$$J_{i+\frac{1}{2}} = \frac{R_i - R_{i+1}}{h} \quad \text{with } R_i = \rho_i(1 - \alpha + \alpha(\rho_{i-1}\rho_{i+1} - \rho_{i-1} - \rho_{i+1} + 1))$$

Our goal here is to prove the limit of this discrete model is the following continuum form:

$$\frac{\partial \rho_i}{\partial t} = \frac{\partial^2 K}{\partial x^2}$$

Where $K(\rho) = \rho(1 - 2\alpha\rho + \alpha\rho^2)$ is defined in the paper with diffusivity $D(\rho) = K'(\rho)$, and the “limit” means $h \rightarrow 0$, i.e. the width of spatial grid approaches to 0, as the time was not been discretized in our model.

Let's start with simplifying the expression of R_i for a little bit:

$$\begin{aligned} R_i &= \rho_i(1 - \alpha + \alpha(\rho_{i-1}\rho_{i+1} - \rho_{i-1} - \rho_{i+1} + 1)) \\ &= \rho_i(1 - \alpha + \alpha\rho_{i-1}\rho_{i+1} - \alpha\rho_{i-1} - \alpha\rho_{i+1} + \alpha) \\ &= \rho_i(1 + \alpha\rho_{i-1}\rho_{i+1} - \alpha\rho_{i-1} - \alpha\rho_{i+1}) \end{aligned}$$

Then, we convert the above equality into the expression of R_i only w.r.t ρ_i by Taylor expansions:

$$\begin{aligned} \rho_{i-1} &= \rho(x_{i-1}) = \rho(x_i - h) = \rho(x_i) - h\rho'(x_i) + \frac{h^2}{2}\rho''(x_i) + O(h^3) \\ &= \rho_i - h\rho'_i + \frac{h^2}{2}\rho''_i + O(h^3) \end{aligned}$$

Notes that ρ' represents $\partial_x \rho_i$, the partial derivative w.r.t space variable x , and similarly for ρ'_i , ρ''_i , etc. The same, for ρ_{i+1} , we have

$$\rho_{i+1} = \rho_i + h\rho'_i + \frac{h^2}{2}\rho''_i + O(h^3)$$

Plug in the discrete expression of R_i , we see

$$\begin{aligned} R_i &= \rho_i \left(1 - 2\alpha\rho_i + \alpha\rho_i^2 + \alpha h^2 (\rho_i \rho_i'' - \rho_i'^2 - \rho_i'') \right) + O(h^3) \\ &= \rho_i (1 - 2\alpha\rho_i + \alpha\rho_i^2) + \alpha h^2 (\rho_i^2 \rho_i'' - \rho_i \rho_i'^2 - \rho_i \rho_i'') + O(h^3) \\ &= K(\rho_i) + h^2 H(\rho_i) + O(h^3) \end{aligned}$$

Where $K(\rho) = \rho(1 - 2\alpha\rho + \alpha\rho^2)$ as we mentioned before, and

$$H(f) := \alpha(f^2 f'' - f f'^2 - f f'')$$

Similarly, we can get

$$\begin{aligned} R_{i-1} &= K(\rho_{i-1}) + h^2 H(\rho_{i-1}) + O(h^3) \\ R_{i+1} &= K(\rho_{i+1}) + h^2 H(\rho_{i+1}) + O(h^3) \end{aligned}$$

Now let's turn back to our discrete conservation equation:

$$\begin{aligned} \frac{\partial \rho_i}{\partial t} &= \frac{J_{i-\frac{1}{2}} - J_{i+\frac{1}{2}}}{h} \\ &= \frac{\frac{R_{i-1} - R_i}{h} - \frac{R_i - R_{i+1}}{h}}{h} \\ &= \frac{1}{h^2} (R_{i-1} + R_{i+1} - 2R_i) \\ &= \frac{1}{h^2} (K(\rho_{i-1}) + K(\rho_{i+1}) - 2K(\rho_i)) + (H(\rho_{i-1}) + H(\rho_{i+1}) - 2H(\rho_i)) + O(h) \\ &= \frac{\frac{K(\rho_{i-1}) - K(\rho_i)}{h} - \frac{K(\rho_i) - K(\rho_{i+1})}{h}}{h} + (H(\rho_{i-1}) + H(\rho_{i+1}) - 2H(\rho_i)) + O(h) \\ &= \quad \quad \quad (\text{part 1}) \quad \quad \quad + \quad \quad \quad (\text{part 2}) \quad \quad \quad + (\text{part 3}) \end{aligned}$$

For part 1 above, simply by finite difference,

$$\lim_{h \rightarrow 0} \frac{\frac{K(\rho_{i-1}) - K(\rho_i)}{h} - \frac{K(\rho_i) - K(\rho_{i+1})}{h}}{h} = \frac{\partial^2 K}{\partial x^2}$$

For part 2:

$$H(\rho_{i-1}) = \alpha(\rho_{i-1}^2 \rho_{i-1}'' - \rho_{i-1} \rho_{i-1}'^2 - \rho_{i-1} \rho_{i-1}'')$$

Again, we hope to get the consistent expressions on both $H(\rho_{i-1})$ and $H(\rho_{i+1})$ w.r.t only ρ_i , in order to see further cancellation. So by Taylor expansion, $\rho_{i-1} = \rho_i + O(h)$, $\rho_{i-1}' = \rho_i' + O(h)$

and $\rho_{i-1}'' = \rho_i'' + O(h)$, we plug them into $H(\rho_{i-1})$:

$$H(\rho_{i-1}) = \alpha(\rho_i^2 \rho_i'' - \rho_i \rho_i'^2 - \rho_i \rho_i'') + O(h^2) = H(\rho_i) + O(h^2)$$

Similarly,

$$H(\rho_{i+1}) = \alpha(\rho_i^2 \rho_i'' - \rho_i \rho_i'^2 - \rho_i \rho_i'') + O(h^2) = H(\rho_i) + O(h^2)$$

So the part 2 becomes

$$H(\rho_{i-1}) + H(\rho_{i+1}) - 2H(\rho_i) = O(h^2) \rightarrow 0 \text{ as } h \rightarrow 0$$

For part 3, obviously $O(h) \rightarrow 0$ as $h \rightarrow 0$.

Hence, when we take the limit $h \rightarrow 0$, both part 2 and part 3 go to 0, only part 1 left. Now we get exactly the desired continuum equation:

$$\frac{\partial \rho_i}{\partial t} = \frac{\partial^2 K}{\partial x^2}$$